



Application of Business Intelligence to Analysis of Factors for the Prevention of Non-communicable Diseases

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Abstract. In Atlántico, Colombia, the Departmental Health Secretariat has been proactive in promoting healthy lifestyles to prevent non-communicable diseases (NCDs), adhering to the strategies outlined in the Health Action Plan (PAS) as recommended by the Ministry of Health's Promotion and Prevention and Epidemiology and Demography Directorates. However, the efficacy of these activities is often hampered by a reliance on external data sources or studies from regions with dissimilar health determinants. This paper highlights the role of business intelligence tools—including dimensional fact models, multidimensional databases, ETL (Extract, Transform and Load) processes, and data visualization dashboards—in enhancing local data analysis, thereby providing valuable insights for public health management and research within Atlántico. By leveraging accurate, localized data, this approach strengthens decision-making processes in NCD prevention. The study aims to ascertain the impact of a healthy diet and physical activity on preventing NCDs among Atlántico's vulnerable populations, utilizing a confirmatory research methodology that employs observation, surveys, and interviews within the constraints of the COVID-19 pandemic. The findings underscore the critical role of diet and exercise in NCD prevention and demonstrate the efficacy of employing digital analytical tools and business intelligence to inform user-centric health interventions. This research lays the groundwork for developing evidence-based, locally tailored NCD prevention strategies, marking a significant advancement in public health initiatives in Atlántico.

Keywords: Non-communicable diseases prevention · Business intelligence in healthcare · Data analysis for public health · Healthy lifestyle interventions · Evidence-based health policy

1 Introduction

In the realm of contemporary organizational dynamics, digital transformation has gained remarkable traction among government and commercial entities on

a global scale [1,2]. In particular, within Colombia, digital transformation has assumed a crucial role in harnessing knowledge through data analysis, ushered in an era of prominence for data science initiatives aimed at tackling productivity challenges and improving decision-making processes [3–5]. In the domain of public health, such initiatives have gained great importance, especially under the demands of a pandemic. The increasing surge in data volume and its multifaceted nature during pandemics amplify the challenge of rapid decision making. Consequently, it becomes imperative to foster technical capacities for efficient data extraction, transformation, and loading, thus preparing the way for subsequent analysis.

The genesis of Business Intelligence (BI) dates back to the 1960s, when it emerged as a conduit for information exchange among corporations. Subsequent decades witnessed its evolution, notably in the 1980s with the development of computer-based decision-making models [6,7]. Gradually, BI metamorphosed into a distinct paradigm, which encompasses service solutions rooted in information technology (IT) [8]. Contemporary BI solutions pivotally prioritize governed data analytics on cohesive and reliable platforms, highlighting their efficacy in information retrieval. These solutions include a variety of processes and methodologies to collect, store and scrutinize data from operational or business activities. The overarching goal is to optimize performance and facilitate strategic, tactical, and operational decisions [9,10]. One notable contender in this field is Microsoft Power BI, a toolkit that facilitates secure and efficient data access, offering substantial advantages to enterprises. Esteemed for its pragmatic, anticipatory, and intelligent system, Microsoft Power BI adeptly translates intricate datasets into visual graphs, dashboards, and reports. Its prowess extends to the generation of presentation ready graphs, synergistically integrated with Microsoft Excel Power Query, an Engine for Data Extraction, Transformation and Loading (ETL) [11].

In this study, we provide a BI model tailored to extract, transform, load, analyze, and visually represent data collected through IT interventions. This model is poised to investigate Non-Communicable Diseases (NCDs) within the vulnerable populace of Colombia's Atlántico Department. Additionally, a series of techniques were used to determine whether a salubrious diet and adequate physical activity constitute exclusively the linchpin for the prevention of NCDs within the region.

Noncommunicable diseases encompass a cluster of diseases that are predominantly due to chronic factors, often necessitating prolonged treatment and care. This category of diseases emerges as the main contributor to global mortality and incapacitation [12]. Annually, Non-Communicable Diseases account for a staggering 41 million deaths, representing 71% of the global toll. The Americas witness approximately 5.5 million NCD-related deaths per year [13]. Among these casualties, 15 million occur in the age bracket of 30 to 69 years, with more than 85% occurring in low- and middle-income economies. The Americas alone witness 2.2 million people succumbing to NCD before reaching 70 years of age. Globally, cardiovascular diseases rule, contributing to 17.9 million deaths, fol-

lowed by Cancer (9.0 million), Respiratory Diseases (3.9 million) and Diabetes (1.6 million) [13]. Unhealthy practices, ranging from excessive alcohol consumption to sedentary lifestyles and inadequate diets, increase vulnerability to NCD-associated fatalities.

Within Colombia, circulatory system diseases were ruled the leading cause of mortality between 2005 and 2014. In particular, ischemic heart disease accounted for 49.30% (293,458) of deaths related to the circulatory system. An investigative report underscores the importance of the Atlántico Department, ranking fifth nationwide in terms of mortality related to NCDs, marked by an increase in NCD morbidity rates [14]. Although the Department of Health has initiated NCD prevention initiatives based on the recommendations of Colombia's Ministry of Health, the imperative to tailor activities to relevant and recent data, taking into account department-specific factors, remains unassailable. Therefore, a convincing rationale underscores the need for NCD studies and analyses, fostering the knowledge relevant to current and prospective inquiries and thus facilitating the planning of health management within the Atlántico Department. Such planning is supported by its own data and analytical models that promote continuous enhancement.

The following sections of this paper are organized as follows: Sect. 2 provides a summary of the cognate research efforts. Section 3 delineates the proposed methodology that drives this investigation. Together, Sect. 4 presents the results and analyses arising from the implementation of the BI model. Finally, Sect. 5 discloses the statement that encompasses the conclusions distilled from this research effort.

2 Related Works

Business intelligence has recently been applied and developed in various systems and environments. Authors of [15] who investigated the critical success factors for the implementation of a BI system by comparing them with those in practice, collected through interviews and questionnaires with experts. The comparison revealed common factors, such as firm management support, a lightweight approach, user acceptance, a project team, and data quality. Furthermore, the interviews exposed additional factors, in particular, the performance of the BI system, the definition of standards, terminology, and key performance indicators; and the institutionalization and integration of BI.

The critical success factors for implementing BI in an Enterprise Resource Planning (ERP) environment have been studied by [16]. The authors used the Decision-Making Trial and Evaluation Laboratory Model (DEMATEL) to analyze information from a strategic cement manufacturing company, evaluating 13 factors related to organization, development processes, technology, and the external environment. Top management support, effective project management, system reliability, flexibility, scalability, and supplier selection are the most relevant factors. Subsequently, Muller *et al.* [17] provided a multidimensional model for analyzing critical success factors for the implementation of Business Analytics. The model was empirically validated using data from 69 food industry

partners. Factors that increase company performance include senior management support, IT infrastructure, and system quality.

The authors in [18] studied the implementation of BI to help higher education organizations make effective decisions, improve teaching and learning services, improve competitiveness, design or produce new strategies and policies. The authors identified a set of factors that influence the use of BI in Malaysian higher education institutions. They found that nine factors, organized into three dimensions (*i.e.*, organizational, process, and technological), promote the use of the BI system and can be utilized by educational institutions. Although the authors argue that the number of successful BI implementations in Higher Education institutions still needs to be increased, their implementation can help institutions obtain crucial patterns and predictions to formulate strategies, generate knowledge, and support decision-making processes.

Janyapoon *et al.* [19] reviewed the evidence on the implementation of BI in the healthcare sector and the success factors contributing to its success. The study included interviews with senior hospital executives and information technology personnel with BI experience in Thailand's healthcare industry. This study aimed to provide evidence on the factors that influence the implementation and success of BI in hospitals. Some studies by Awawdeh *et al.* [20] have used BI to test and evaluate the impact of digital entrepreneurship on digital supply chains in Jordanian hotels. The study was tested by designing questionnaires to collect relevant information from the supply chain. The results demonstrate the importance and impact of BI and its applications in digital entrepreneurship.

- Identifying high-risk individuals: BI can be used to identify people at high risk for developing NCD. This information can be used to target prevention interventions to those who are most likely to benefit.
- Tracking patient progress: BI can be used to track patient progress over time. This information can be used to identify patients who do not respond to treatment and to adjust treatment plans as needed.
- Evaluating prevention interventions: BI can be used to assess the effectiveness of prevention interventions. This information can be used to improve the design and implementation of future prevention programs.

Finally, Viteri *et al.* [21] introduced a methodology to develop BI solutions in the banking sector by studying the critical success factors and approaches proposed in the literature. The objective was to improve decision making in a business unit that works with internet banking companies, in order to reduce decision-making time, personnel, and costs. Additionally, tools were used to collect and record data on time, personnel, and expenses associated with decision making.

This study proposes exploring, visualizing and analyzing information regarding NCDs in the Atlántico Department (Colombia) through BI using empirically verified confirmatory research at the integrative level. Furthermore, it investigates whether a healthy diet and adequate physical activity are the only significant factors in preventing NCDs among the vulnerable population of the Department.

3 Methodology

Taking into account the restrictive biosafety regulations due to COVID-19, the observation technique was used to observe the public health agents of the community in the Atlántico department, including community leaders, mothers, religious groups, recreation groups, and sports groups. This technique is supplemented by interviewing the Coordinator of the Healthy Lifestyles dimension and his team.

The sample consisted of 303 vulnerable people aged 18 years. In this research, the term “vulnerable” refers to people in a state of multidimensional poverty defined by the Technical Management of Methodology and Statistical Production of the National Administrative Department of Statistics in Colombia. The Respondent-Driven Sampling (RDS) method was used to randomly select the population and sample. The citizen survey was conducted digitally through a Web application (<https://evsatlantico.com/>) from August 1 to November 12, 2021. By thorough analysis of the current NCD prevention model in the vulnerable population of the Department of Atlántico (Colombia) and research of the diet and physical activity approach, the objective was to prevent these diseases by generating knowledge among the population and public health agents through the application of the BI concept, including its methodology and data visualization tools.

It should be noted that the implementation of the Comprehensive Health Care Routes (RIAS) model by the Colombian Ministry of Health is integral to this research. This model defines the guidelines to be followed in the management of public health for citizens in the department of Atlántico (COMMUNITY) and the HEALTH SYSTEM AGENTS: (municipal health secretaries, E.S.E, IPS), with the following actions defined for each actor:

- COMMUNITY: To raise awareness about the timely identification of health risk factors for each Atlántican, encouraging efficient use of health resources. “Community health.”
- HEALTH SYSTEM AGENTS: Highlight the comprehensive provision of health services in the realm of NCD prevention carried out by EAPBs, ESEs, and local health departments within the RIAS framework.
- DEPARTMENTAL HEALTH SECRETARY: Design indicators to validate factors that influence and impact prevention of noncommunicable diseases in the vulnerable population of the Atlántico department.

It is also necessary to have a phrase about the concept of “prevention of the vulnerable population of the department of Atlántico” as a guiding principle in the application of the business intelligence model, defining it as follows: “It is the process by which a citizen becomes involved in different aspects that mitigate their risk factors for developing non-communicable chronic diseases, committing to comply with the norms outlined in the Comprehensive Health Care Routes of the Ministry of Health, MSPS,” allowing the identification of the phrase in two dimensions:

- Dimension 1: Identification of common risk factors.
- Dimension 2: Commitment.

3.1 Business Intelligence Methodology

This research proposes and implements a BI methodology based on the following seven elements:

1. Understanding of the business: This element includes identifying symptoms and problems, defining the Business Success Criteria and BI success criteria, and assessing the situation with respect to resources, assumptions, risks, and cost-benefit analysis.
2. Understanding of data: This element includes collecting initial data and validating infrastructure and security.
3. Prototyping: This element covers identifying User Stories and determining the type of interface required.
4. Data preparation: This element involves designing the architecture of the solution, executing the architecture design, and creating a report on data quality.
5. Modeling: This element involves selecting a modeling technique, generating a test plan, building the model, and evaluating it.
6. Evaluation: This element comprises performing the usability test, evaluating the results, reviewing the process, and determining the next steps.
7. Implementation: This element combines an implementation plan, developing a monitoring and maintenance plan, preparing a final report, performing a project review, analyzing lessons learned, providing incident support, and facilitating continuous improvement.

This methodology was applied to systematize the data, with the aim of implementing statistical techniques for information analysis and ensuring that a fast and high quality analysis is provided to the Department of Health Secretary. A collaborative information model was also designed and implemented between population and public health agents. This model takes information from citizens and health agents in three directions.

1. Increase awareness of the timely identification of risk factors in citizen health so that they can use health resources efficiently (community health).
2. Comprehensive provision of health services in the prevention of NCDs.
3. Design indicators to validate incidence factors and their impact on the prevention of NCDs in the vulnerable population of the Atlántico Department of Health.

The digital tool was designed to capture sociodemographic and anthropometric data, as well as data related to cardiovascular disease, diabetes, chronic and obstructive pulmonary diseases, chronic kidney diseases, cancers, and habits. Once all questions have been completed, the web application (<https://evsatlantico.com/>) generates a document with results based on the questions. The output measures the risk of developing each NCD (see the results in the next section).

4 Results

This section presents the results of the implementation of the BI methodology in its seven elements. This part of the process is essential because the data and information entered by the respondents must be processed, analyzed, and visualized through tools that enable informed decision-making over time.

4.1 Business Understanding

- Identification of symptoms and problems: A set of questions is established to identify symptoms and problems (see Fig. 1).
- The criteria for business success: An analysis of factors that contribute to the prevention of NCD was carried out among the vulnerable population of the Atlántico Department.
- Success criteria for BI success: 303 high-quality data points were collected between August and November 2021.
- Situation assessment: the user who answered the questionnaire with a Microsoft license used the Power BI tool.

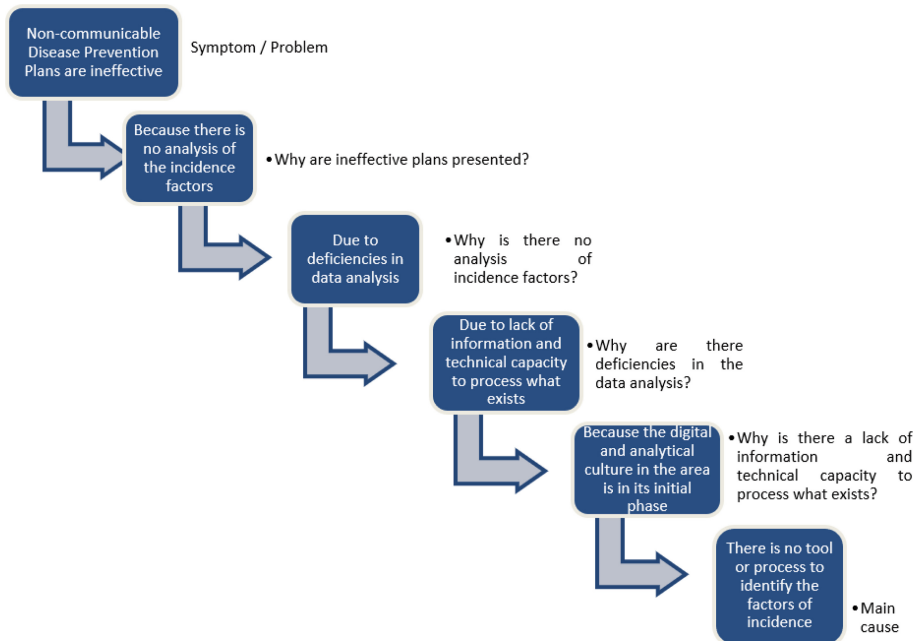


Fig. 1. A scheme to identify symptoms and problems.

4.2 Data Understanding

Table 1 summarizes the components for understanding the information in a model table. Upon analyzing the data to be processed, it was identified that the type of data was structured, indicating its consistency and persistence. The data are in the public domain, as the context of the study is public health and the objective is to describe the factors that initiate NCD through questions. However, data collection was challenging due to restrictions from the COVID-19 pandemic. Therefore, data were stored as text in a repository on a web resource, managed through software capable of handling a maximum delay rate of one minute and a data loss tolerance of 98%.

Table 1. Components for understanding information.

Data Types	Domains	Data Volume	Data Rate	Data Variety	Data Veracity
Structured	Public	Small	1 min. delay	Text only	Tolerance 98%
Data Value	Focus	Storage	Processing	Objective	Product
Not applicable	Past and present	Central server (Web)	Database consultation	Describe reality	Dashboard report and metrics

4.3 Prototyping

This element considers the user’s information needs and the interface type. Information needs include the following.

- User history.
- How: Team in the dimension area of healthy lifestyles.
- I want: A dashboard that allows us to visualize tobacco consumption, healthy diet, physical activity, alcohol consumption, attendance at doctor follow-ups and controls, medications prescribed for hypertension, Findrisc score, and survey traceability, sorted by time, sex, and life cycle.
- For: Analyzing factors to prevent NCD of vulnerable population in NCD in (Colombia).
- Type of interface: A dynamic dashboard with filters and a map of municipalities in the Atlántico Department.

4.4 Data Preparation

Architecture. It is relevant to the design and execution of solutions. This study implemented a multidimensional model of the constellation of facts based on the “KNOW YOUR HEALTH RISK” survey questions and the user story. The model consisted of 11 fact tables and four-dimensional tables. Six fact tables group data on people with a history of disease, as described below.

- *fact_{history}HTA*

- *fact_antecedentesCancer*
- *fact_antecedesDiabetes*
- *fact_antecedentesECR*
- *fact_historyEPOC*
- *fact_historyofOtherdiseases*

In addition, five fact tables grouped the questions used to identify factors related to disease prevention.

- *fact_CouldBeCOPD*
- *fact_FINDRISC*
- *fact_habits*
- *fact_hhealthattention*
- *fact_survey*

There are also four dimensions, which include.

- *respondent_dim* (person's information)
- *dim_nneighborhood*
- *dim_municipality* (information to geolocate the person)
- *dim_surveyor* (information of the surveyor)

The architecture used in this study involves connecting the data source layer through the Power BI desktop interface, as shown in Fig. 2. In this case, the data source was the remote MySQL (Web) database. After the connection, the extraction, transformation, and load processes were carried out using Power query, which is a Power BI tool, to build the multidimensional model. Once validated, it is passed on to the visualization and analytical layer to build the requested dashboard in the user story.

Data Quality. The fields used to check the quality of the data included the survey ID, *theinterviewer_ID*, *therespondent_ID* and the survey date (see Fig. 3).

The modeling, evaluation, and implementation elements were developed within the architecture used, where Microsoft Power BI is remotely connected to the data source layer (in this case, the MySQL database hosted on the Web). After connecting, data are extracted, transformed, and loaded through a Power query, a Power BI tool, and a multidimensional model is built. Once the model is validated with metrics such as valid data (100%), error data (0%), and empty data (0%), it is passed to the visualization and analytical layer, where the requested dashboard is created from the user story.

The results of the instrument's application enable municipal and departmental health secretaries to gain visibility on the comprehensive provision of health services in NCD prevention. Furthermore, it allows them to observe the incidence factors and their impact on the prevention of NCDs in the vulnerable population of the Atlántico Department. We can draw the following partial conclusions based on the results of the analyzed and visualized sample (303 individuals from the municipalities of Barranquilla, Galapa, Soledad, and Baranoa) using the Power BI tool.

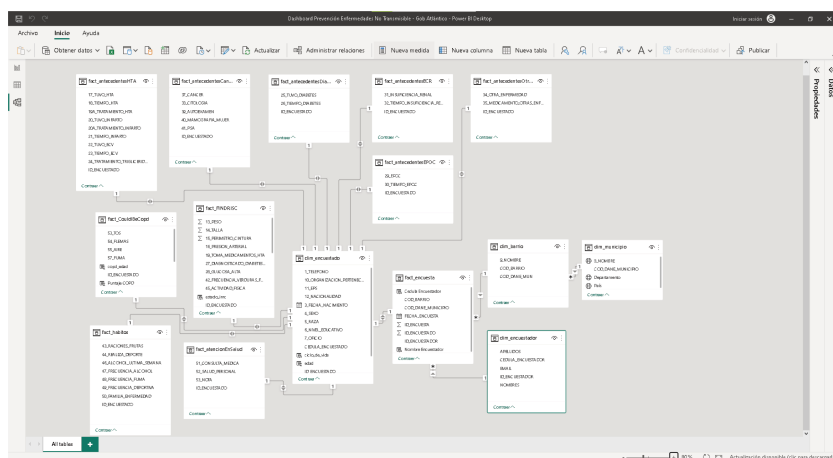


Fig. 2. Design structure model.

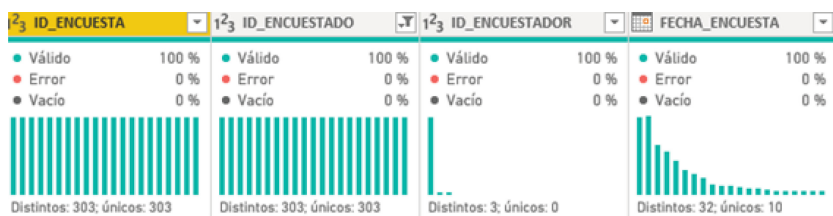


Fig. 3. Design of the Dashboard.

- **Life cycle - old age:** The observed data indicate a high risk of diabetes due to a FINDRISC score greater than 12 points. Globally 70% people do not exercise and 75% do not have a healthy diet (see Fig. 4).
- **Life cycle - adulthood:** The results show a high risk of diabetes due to a FINDRISC score greater than 12 points. Globally, 74% of people do not exercise and 74% do not have a healthy diet (see Fig. 5).
- **Life cycle - youth:** The observed data reveal that young people are at high risk for diabetes due to a FINDRISC score greater than 12 points. Globally, 77% of people do not exercise and 65% do not have a healthy diet (see Fig. 6).
- **Life cycle - adolescence:** In the case of adolescents, there is a high risk of diabetes due to a FINDRISC score greater than 12 points. Globally, 75% of people do not exercise and 50% do not have a healthy diet (see Fig. 7).

This panorama enables us to begin to understand and study NCDs. Additionally, we observe a percentage trend that allows us to identify the absence of a healthy diet and physical activity among the surveyed individuals. Based on the preceding analysis, it can be inferred that prevention of NCDs in the Department of Atlántico occurs when indicators exhibit the following values (see Table 2).

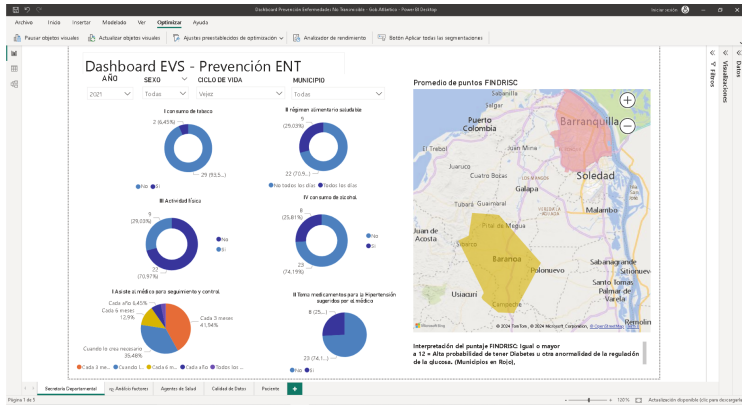


Fig. 4. Results for Life cycle - old age.

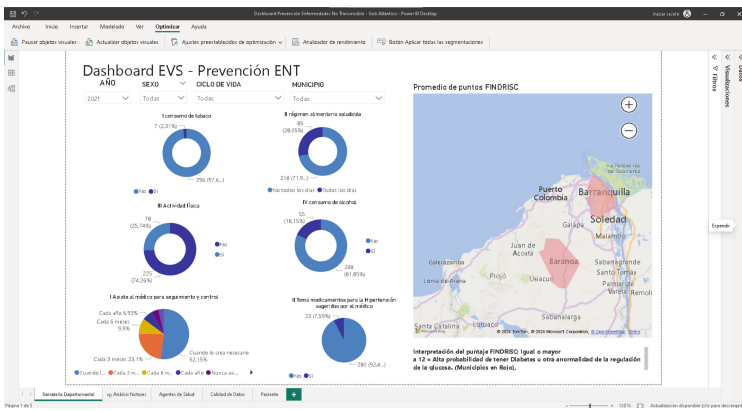


Fig. 5. Results for Life cycle - adulthood.

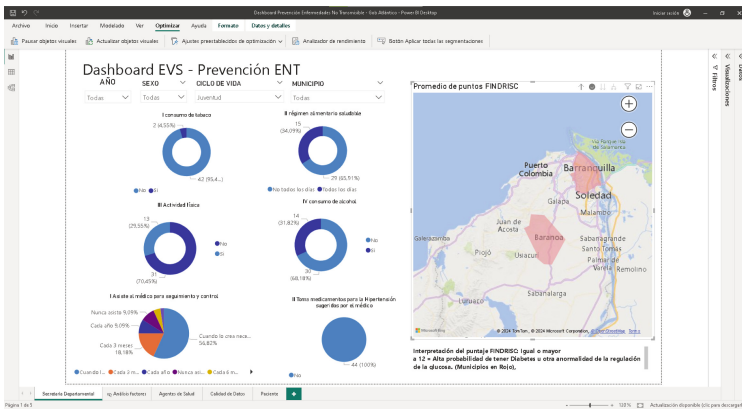


Fig. 6. Results for Life cycle - youth.

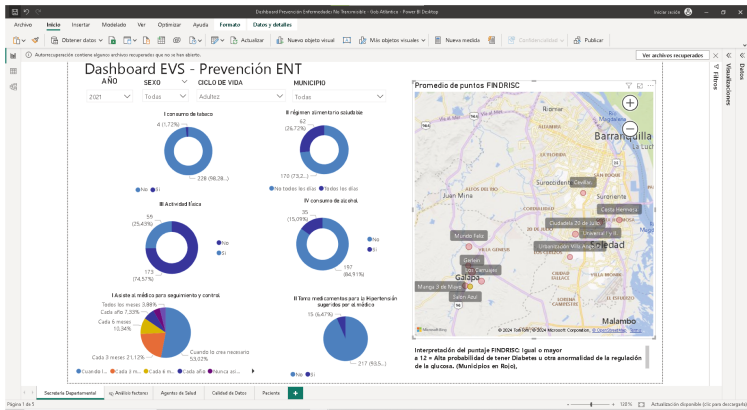


Fig. 7. Results for Life cycle - adolescence.

Table 2. Dashboard of indicators for NCD prevention.

Dimension	Indicator	Reference value
Identification of common risk factors	Use tobacco	No
	Healthy diet	Every day
	Physical activity	Yes
	Alcohol consumption	No
Commitment	Go to the doctor for follow-up and control	Every month, Every 3 months, Every 6 months, Every year
	Take medications for hypertension suggested by the doctor	Yes

5 Conclusions

In the Atlántico Department in Colombia, efforts toward digital transformation have begun to analyze risk factors that affect the prevention of NCD. The study focused on the dimension of a healthy lifestyle, including the analysis of factors such as healthy diets and adequate physical activity to prevent NCDs in the vulnerable population; A BI tool was designed and implemented to analyze risk factors for the prevention of NCDs. This tool was built with the digital instrument “KNOW YOURHEALTH RISK”, which was applied for the first time in the Department, ensuring the quality of captured data.

Using this tool has facilitated the analysis of variables that impact the prevention process in citizens of the Atlántico Department. Factors such as healthy diets and physical activity were found to be essential in the prevention of NCD. However, other factors, such as alcohol consumption, timely medical care, and medication use, must be considered. The business intelligence tool developed for data analysis and visualization was user-centered, allowing for generating infor-

mation from data obtained from the digital survey. The tool provided a practical and easy-to-use interface for data modeling.

In the future, further investigations are expected to be conducted. First, extension through new samples from citizens versus samples taken by health service providers, and second, quarterly evaluation of the performance and building analysis of new data generated by the Business Intelligence tool and the digital instrument “KNOW YOURHEALTH RISK”.

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